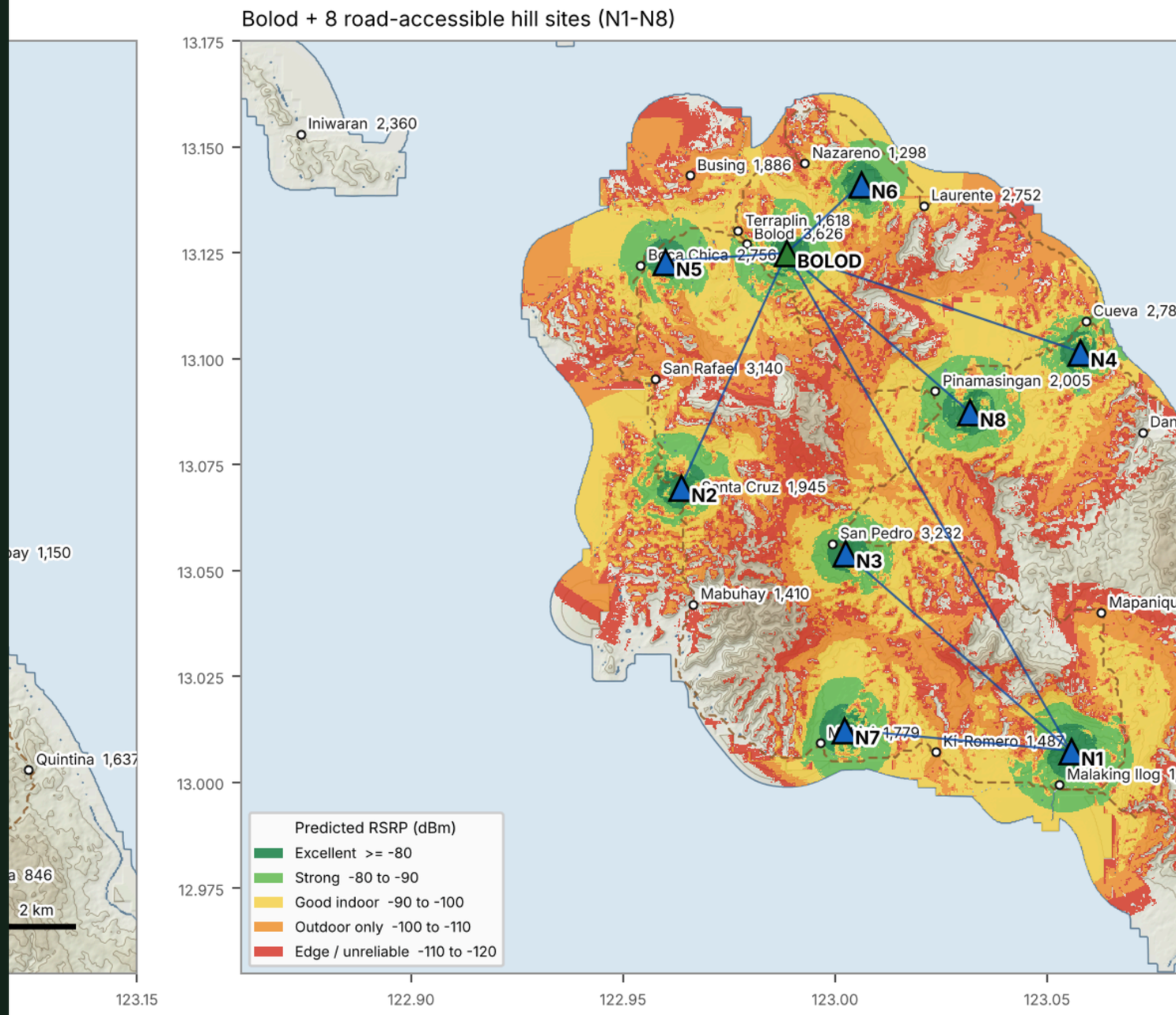
TECHNICAL ADVISOR'S REVIEW · SANGGUNIANG BAYAN OF SAN PASCUAL, MASBATE

San Pascual Cell-Site Review

The proposed tower in deeper Barangay Bolod near the Pinamasingan Road, tested against the existing Bolod hill towers and against road-accessible ridge sites, using a coverage model calibrated to what residents actually receive.

Revision 2 · October 2026 · Field-calibrated coverage model (usable range 3–4 km)



SUMMARY

The proposed tower copies Bolod. Ridge sites reach new people.

36,021
residents in 18 barangays with no usable signal (81% of 44,449)

87%
of the contractor site's coverage is already covered by Bolod

0
unserved residents it adds in all three location scenarios with a 40 m mast (at most 15 even at 60 m)

4,314
added by the best single ridge site (Malaking Ilog–Mapanique)

What we found

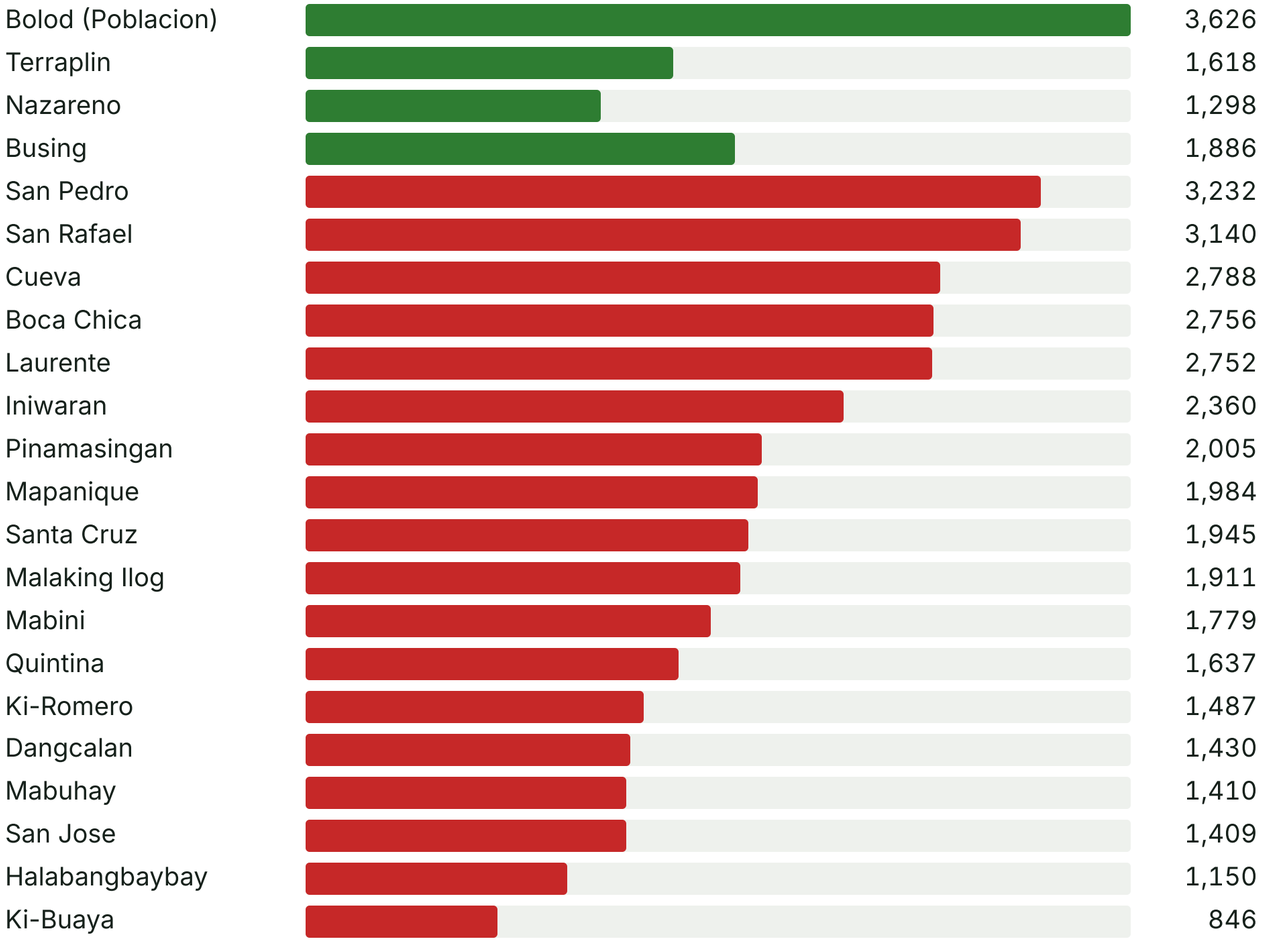
- The existing Bolod towers reach about 3–4 km. They serve Bolod, Terraplin, Nazareno and Busing, and do not reach Laurente. Our model now reproduces this.
- The contractor's antennas would sit 46 m lower than Bolod's, 1–2 km away. Everything they can reach, Bolod already reaches.
- Because signal only travels 3–4 km here, the unserved barangays need towers near them, on high ground. Every ridge site we propose is within 1 km of a road and has a clear microwave link back.

What we recommend

- **Do not endorse the deeper-Bolod site as a coverage project.**
- Ask for the Malaking Ilog–Mapanique and San Rafael–Santa Cruz ridges first: +8,403 residents, taking service from 20% to 39%.
- If the contractor will only build on its own road: the Pinamasingan ridge (+2,615) or the Cueva ridge (+2,813).
- Ask for an eight-site roadmap: 71% of residents within service.

THE PROBLEM

36,021 of 44,449 residents have no usable signal



Served by Bolod hill today (field report)

Not served

Four barangays around the poblacion and port are served: **8,428** people (19%).

The other eighteen, **36,021** people, sit behind the hills or beyond the towers' 3–4 km reach. The largest are San Pedro (3,232), San Rafael (3,140), Cueva (2,788), Boca Chica (2,756) and Laurente (2,752).

Source: PSA 2020 Census via PhilAtlas. The 2024 POPCEN total is 44,641 (+0.4%).

What changed in this revision

1. Laurente

The first map showed Bolod reaching parts of Laurente. Field reports say it does not. The terrain confirms it: a spur rises about 41 m above the radio path, 3.7 km out.

Now: Laurente **1%** covered.

2. Range

The first model used a low-band, open-terrain budget and let signal run 10 km or more. Residents report usable signal ends at about 3–4 km.

Now: median land range **3.2 km** (most directions 2.8–3.45 km).

3. The map itself

The first maps used one flat colour over a grey relief, which read like a flood-depth map and hid how strong the signal is.

Now: graded RSRP classes from excellent to edge, on a contour basemap with 1–4 km range rings, land only.

All conclusions were recomputed. The direction of the finding is unchanged and stronger: the contractor's site adds **zero** unserved residents. Because each tower now reaches less, more towers are needed, and each must sit close to the people it serves.

Calibrated to what residents receive from Bolod hill

BARANGAY	KM	FIELD REPORT	USABLE	INDOOR	CHECK
Bolod (Poblacion)	1.1	Served	87%	72%	✓ matches
Terraplin	1.4	Served	77%	47%	✓ matches
Nazareno	2.4	Served	55%	18%	✓ matches
Busing	3.2	Served	64%	9%	✓ matches
Boca Chica	3.8	Not served	17%	0%	✓ matches
San Rafael	4.7	Not served	0%	0%	✓ matches
Pinamasingan	5.3	Not served	0%	0%	✓ matches
Laurente	3.7	Not served	1%	0%	✓ matches

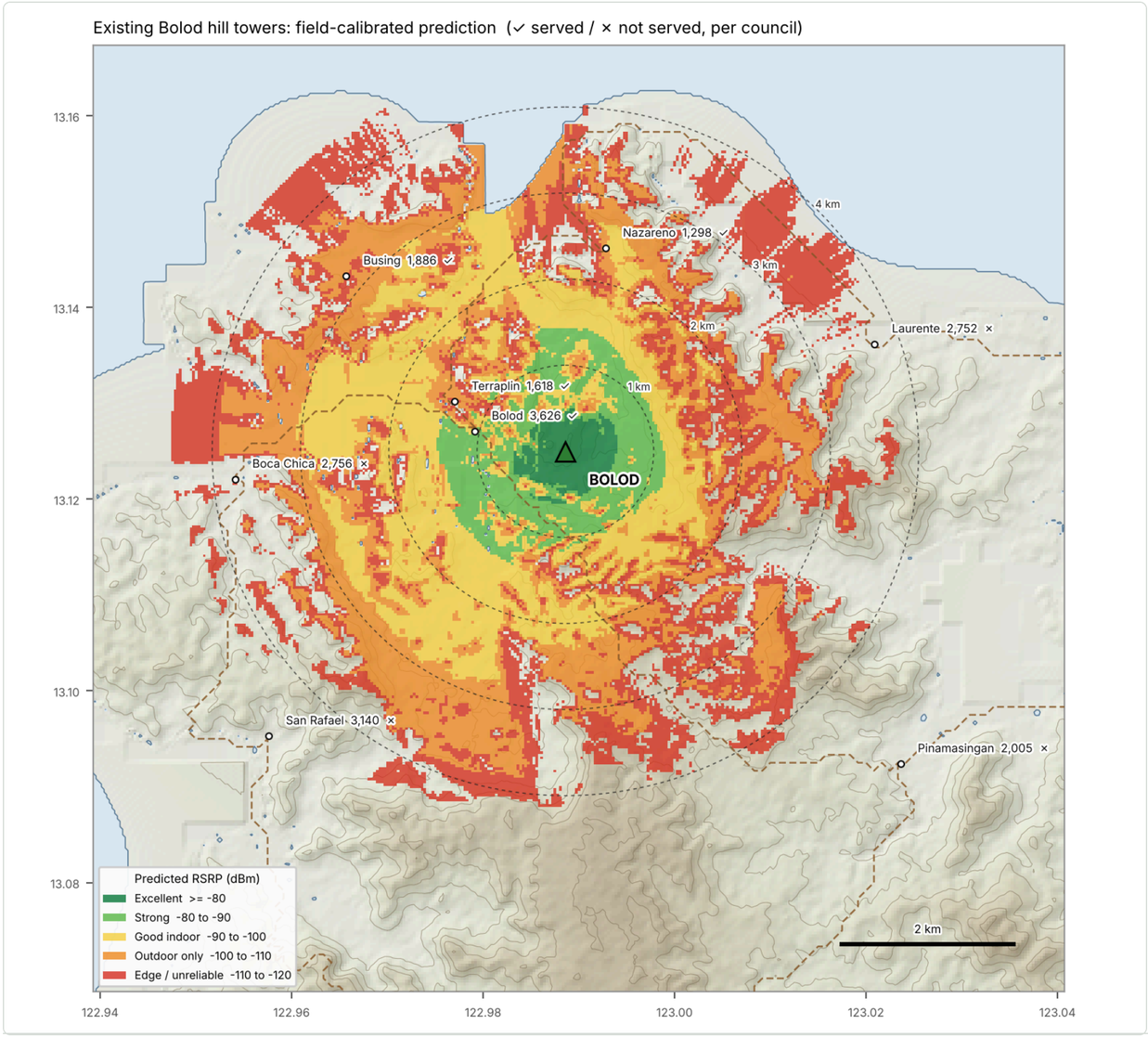
Usable = share of the barangay's settled area at RSRP ≥ −110 dBm; indoor = ≥ −100 dBm. Barangays beyond 5.5 km all predict 0%.

The model

- Terrain: SRTM 30 m elevation, radar spikes removed.
- Path loss: COST-231 Hata (suburban) at 1800 MHz, plus terrain diffraction over the actual ridges (Deygout, ITU-R P.526).
- Effective antenna height above each receiver, so hilltop sites get their real advantage.
- Handset uplink fade beyond 3.5 km, matching the observed 3–4 km reach.
- Thresholds: ≥ −110 dBm usable outdoors, ≥ −100 dBm indoors.

The free parameters were tuned so Bolod hill covers its four barangays and nothing else, ending at 3–4 km. A drive test should replace this tuning before final siting.

Existing Bolod hill towers: predicted signal strength



Signal is strong around the poblacion and port, and fades to edge level by 3–4 km.

Busing and Nazareno sit in the outdoor-only and edge bands, consistent with "signal, but weak indoors".

Laurente (2,752) is behind a spur and gets nothing. **Boca Chica** gets edge signal on its eastern side only.

Pinamasingan and San Rafael are beyond reach.

Reading the colours

Excellent

`≥ −80 dBm`

Full speed indoors and out. Video calls work everywhere.

Strong

`−80 to −90`

Reliable inside concrete houses. Good data.

Good indoor

`−90 to −100`

Calls and messaging indoors; data slows in back rooms.

Outdoor only

`−100 to −110`

Works outside or by a window. Drops indoors. Counted as "usable".

Edge

`−110 to −120`

Signal bars appear but calls drop. Not counted as served.

No colour

`< −120`

No service.

RSRP (Reference Signal Received Power) is the LTE signal level a phone measures. It is the figure the contractor's own propagation map should show, and what a drive test records.

CONTRACTOR'S SITE

Where the contractor wants to build

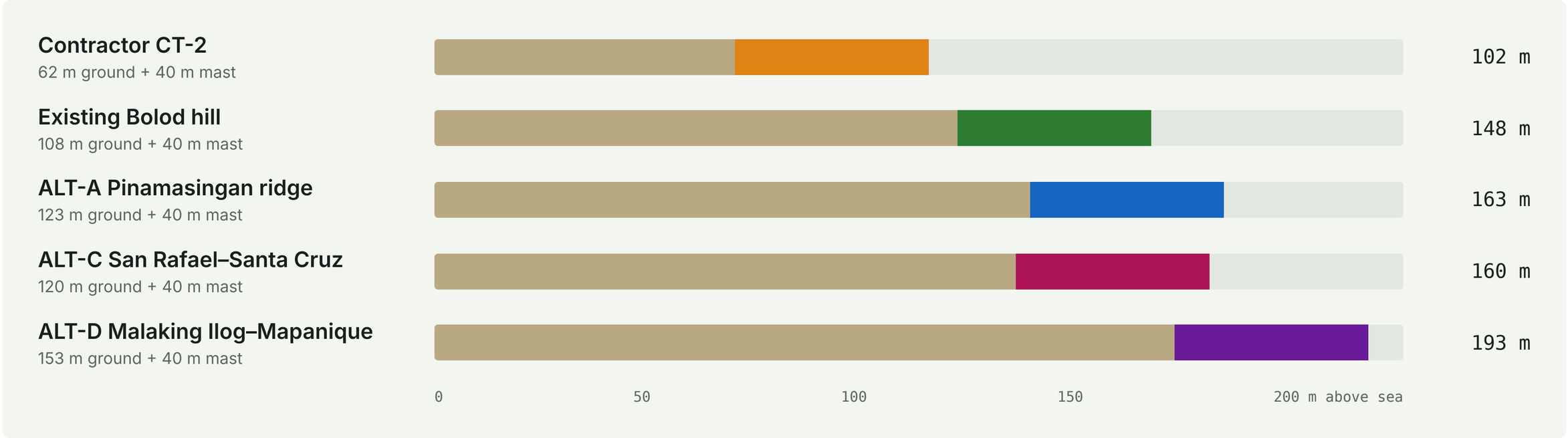
The application places the tower in deeper Barangay Bolod near the Pinamasingan Road. We did not have exact coordinates, so we tested three plausible spots on that road, 1–2 km south of the existing towers.

SCENARIO	LAT, LON	GROUND	ANTENNA TOP (40 M MAST)	FROM EXISTING TOWERS	AREA ALREADY COVERED BY BOLOD	UNSERVED RESIDENTS ADDED	ADDED AT 30 / 45 / 60 M MAST
CT-1	13.1158, 122.9892	60 m	100 m	1.0 km	94%	0	0 / 0 / 0
CT-2	13.1100, 122.9908	62 m	102 m	1.7 km	87%	0	0 / 0 / 0
CT-3	13.1067, 122.9933	73 m	113 m	2.1 km	75%	0	0 / 0 / 15
Bolod hill (existing)	**13.1250, 122.9886**	**108 m**	**148 m**	—	—	—	—

Whichever spot it is, and however tall the mast, the result is the same: **no unserved barangay gains usable service.**

CONTRACTOR'S SITE

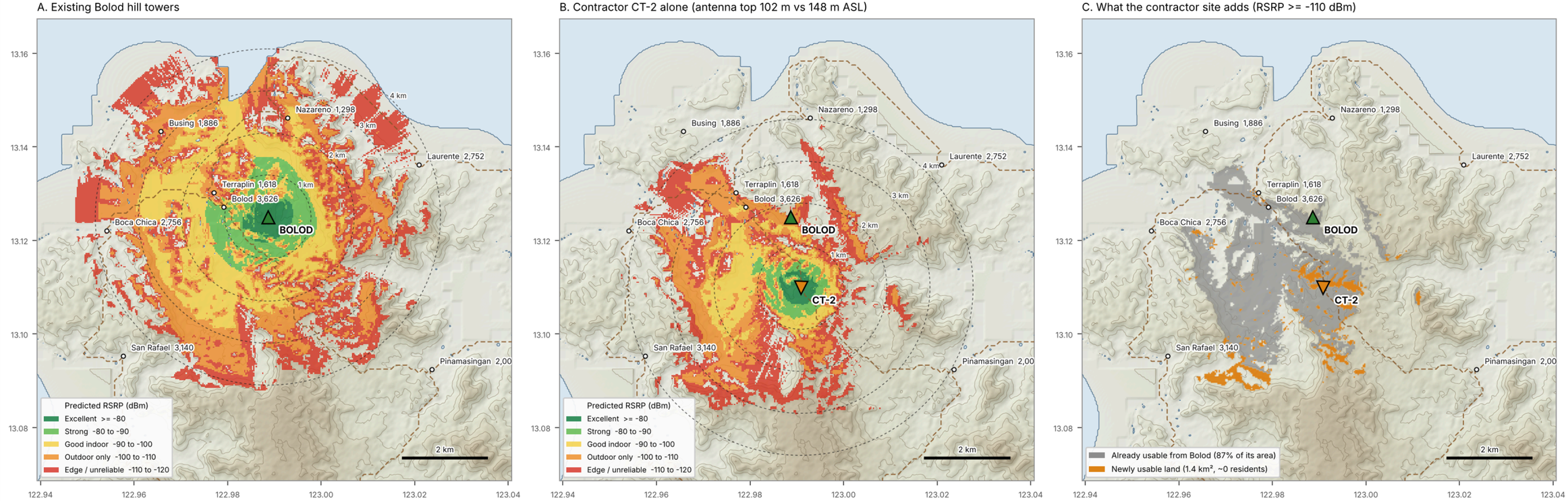
A lower antenna next to a higher one sees nothing new



Drawn to scale. The contractor's antennas would be 46 m below the antennas already on Bolod hill. Any ridge that blocks Bolod also blocks them. The ridge alternatives put antennas 12–45 m higher than Bolod's, and next to the people they serve.

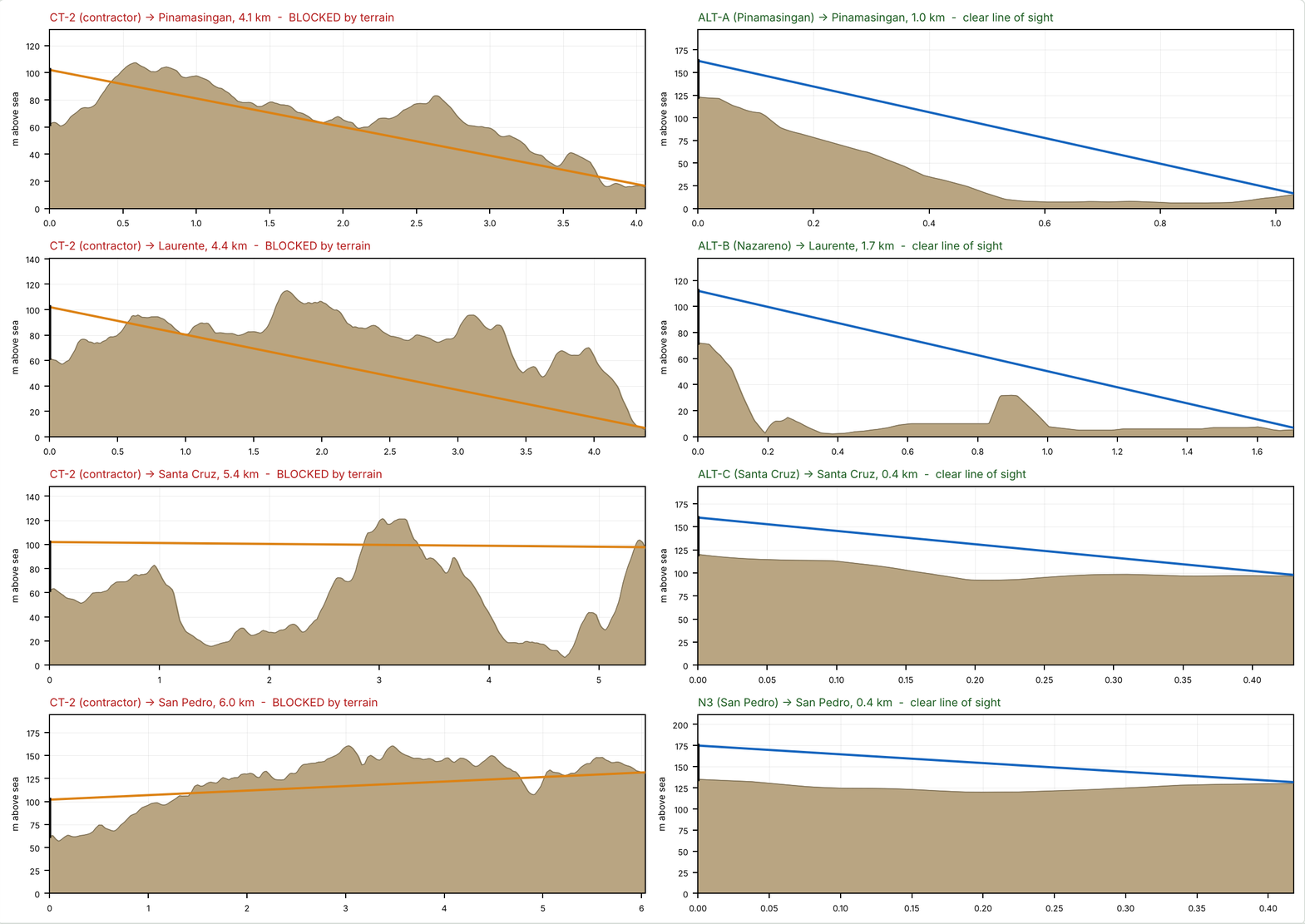
CONTRACTOR'S SITE

87% of its coverage is already covered



C shows the only land the contractor's site would add: 1.4 km² of slopes and gullies south of the towers, with no barangay centre in it. Unserved residents gained: **0**.

The terrain between the contractor's site and the unserved barangays



Left: from the contractor site. Right: from the nearest proposed ridge site. Brown is the ground; the line is the direct radio path to a phone 1.5 m above ground.

Pinamasingan (4.1 km): a spur next to the site blocks the path.

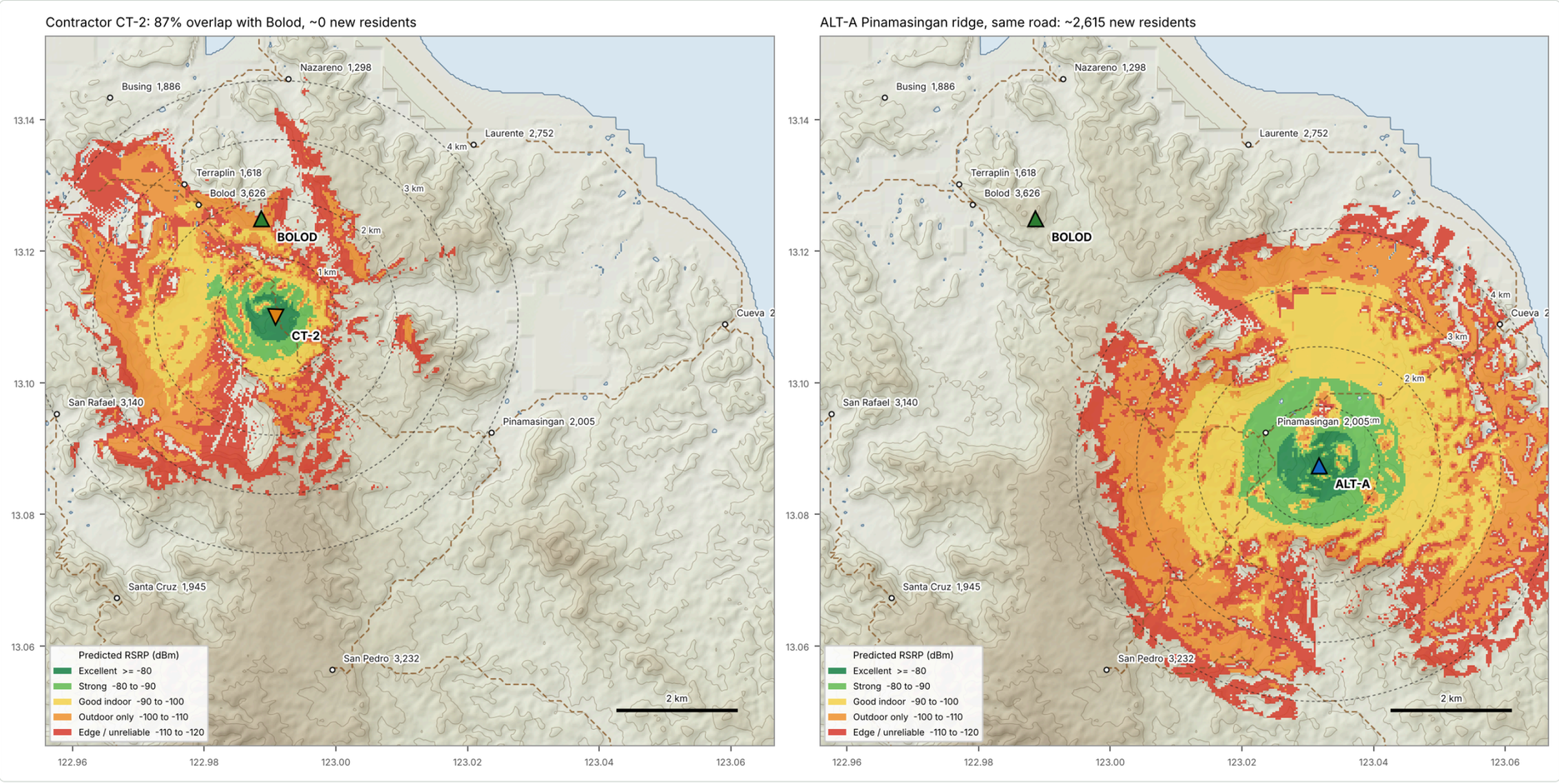
Laurente (4.4 km): the path runs under 3 km of hills up to 115 m.

Santa Cruz (5.4 km) and **San Pedro** (6.0 km): beyond reach and behind the 160 m central ridge.

From the ridge sites, each of these barangays has clear line of sight.

BETTER COMPROMISE

Same road, higher ground



ALT-A sits on the same Poblacion–Pinamasingan road, about 4.5 km past the contractor's spot, with a 723 m access spur. It covers Pinamasingan fully, and part of Cueva, for **+2,615** residents, against zero for the proposed spot.

BETTER COMPROMISE

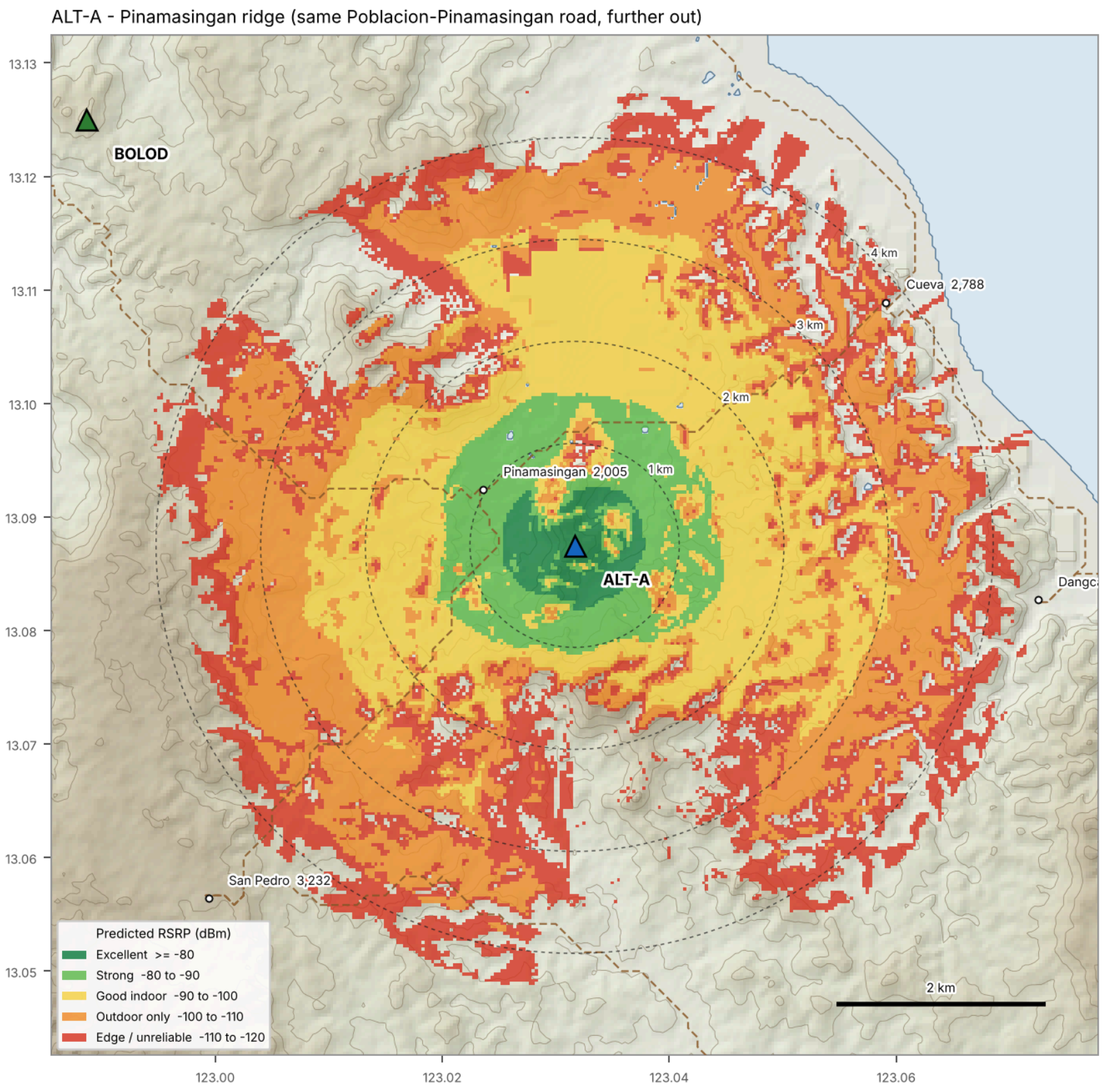
Five ridge sites the contractor can still drive to

SITE	LAT, LON	GROUND	ROAD SPUR	MICROWAVE TO BOLOD	NEW RESIDENTS (USABLE)	(INDOOR)	BARANGAYS LIFTED (SHARE OF SETTLED AREA)
ALT-A Pinamasingan ridge same Poblacion-Pinamasingan road, further out	13.0875, 123.0317	123 m	723 m	6.3 km · 40 m	**2,615**	1,947	Pinamasingan 100%, Cueva 22%
ALT-B Laurente-Nazareno ridge north-coast road to Laurente	13.1414, 123.0061	72 m	971 m	2.6 km · 20 m	**2,293**	1,914	Laurente 85%, Nazareno 38%
ALT-C San Rafael-Santa Cruz ridge west-coast road via San Rafael	13.0697, 122.9636	120 m	43 m	6.7 km · 15 m	**4,089**	2,401	Santa Cruz 99%, San Rafael 62%, Mabuhay 14%
ALT-D Malaking Ilog-Mapanique ridge inland Pinamasingan-San Pedro-Mapanique road	13.0075, 123.0556	153 m	985 m	15.0 km · 35 m	**4,314**	2,323	Malaking Ilog 99%, Mapanique 64%, Ki-Romero 62%, San Jose 16%
ALT-E Cueva ridge Poblacion-Quintina road via Cueva	13.1017, 123.0578	62 m	410 m	7.9 km · 35 m	**2,813**	2,069	Cueva 91%, Pinamasingan 12%
CT-2 Contractor, deeper Bolod	13.1100, 122.9908	62 m	0 m	1.7 km	**0**	0	none

Screened from 93 hilltops within 1 km of a road. Each has a microwave path to Bolod clearing 60% of the first Fresnel zone (and 30% under sub-refraction) at 15 GHz, with masts of 45 m or less at both ends.

BETTER COMPROMISE

ALT-A · Pinamasingan ridge

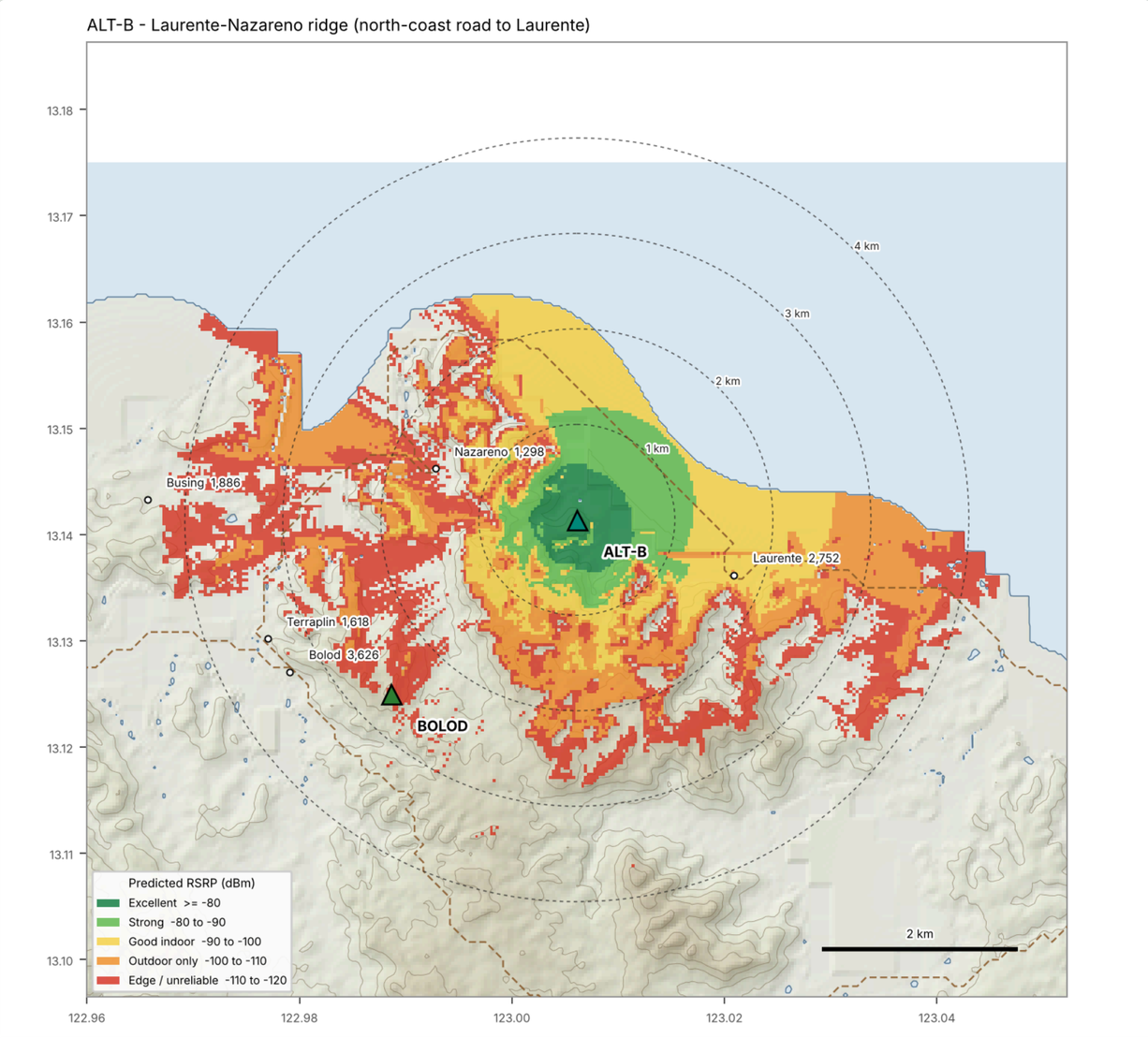


The direct answer to the contractor's road. It stays on the Poblacion–Pinamasingan road and climbs a 123 m ridge above Pinamasingan.

LOCATION	13.0875 N, 123.0317 E
GROUND / ANTENNA TOP	123 m / 163 m above sea
ROAD ACCESS	723 m spur · same Poblacion-Pinamasingan road, further out
BACKHAUL	6.3 km to Bolod · 40 m masts at 15 GHz (45 m at 7 GHz)
NEW RESIDENTS	**2,615** usable · 1,947 indoor-grade
OVERLAP WITH BOLOD	0%
BARANGAYS LIFTED	Pinamasingan 100%, Cueva 22%

BETTER COMPROMISE

ALT-B · Laurente-Nazareno ridge

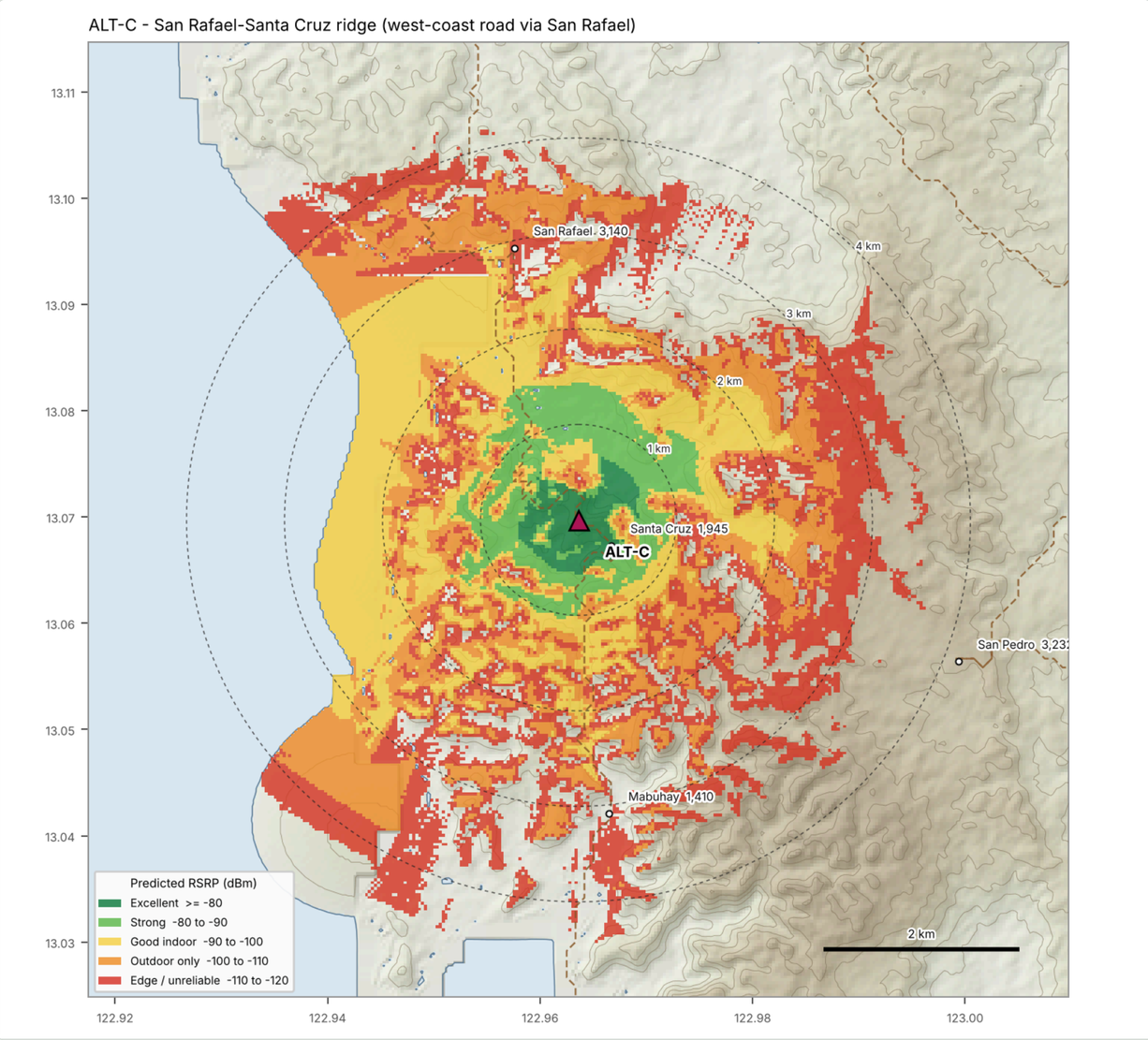


A 72 m knoll on the north-coast road between Nazareno and Laurente. It reaches Laurente, which Bolod cannot, with a short 2.6 km hop.

LOCATION	13.1414 N, 123.0061 E
GROUND / ANTENNA TOP	72 m / 112 m above sea
ROAD ACCESS	971 m spur · north-coast road to Laurente
BACKHAUL	2.6 km to Bolod · 20 m masts at 15 GHz (20 m at 7 GHz)
NEW RESIDENTS	**2,293** usable · 1,914 indoor-grade
OVERLAP WITH BOLOD	22%
BARANGAYS LIFTED	Laurente 85%, Nazareno 38%

BETTER COMPROMISE

ALT-C · San Rafael-Santa Cruz ridge

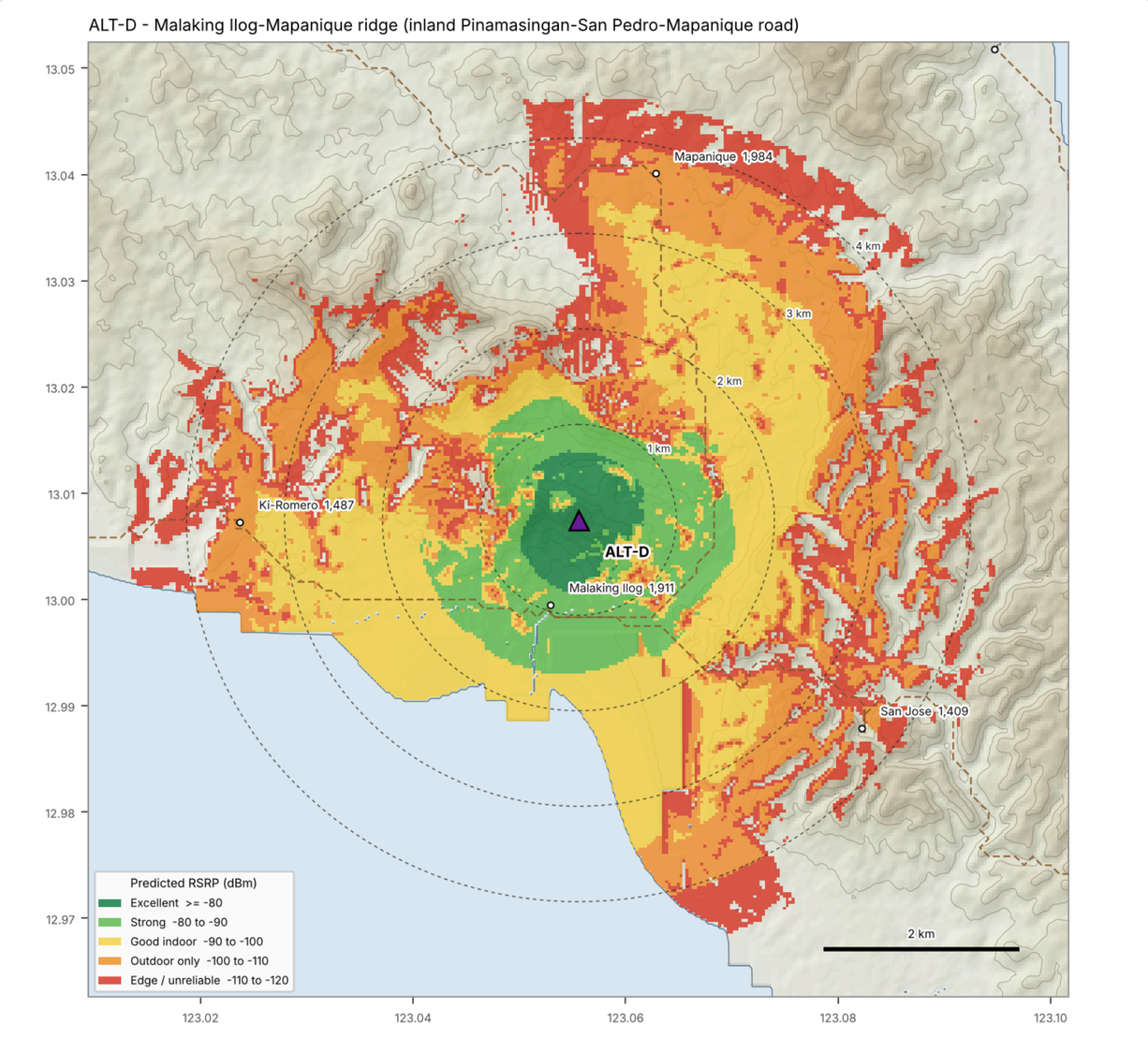


On the west-coast road itself (43 m from it). It covers Santa Cruz and most of San Rafael, with the easiest backhaul on the island.

LOCATION	`13.0697 N, 122.9636 E`
GROUND / ANTENNA TOP	120 m / 160 m above sea
ROAD ACCESS	43 m spur · west-coast road via San Rafael
BACKHAUL	6.7 km to Bolod · 15 m masts at 15 GHz (15 m at 7 GHz)
NEW RESIDENTS	**4,089** usable · 2,401 indoor-grade
OVERLAP WITH BOLOD	0%
BARANGAYS LIFTED	Santa Cruz 99%, San Rafael 62%, Mabuhay 14%

BETTER COMPROMISE

ALT-D · Malaking Ilog-Mapanique ridge

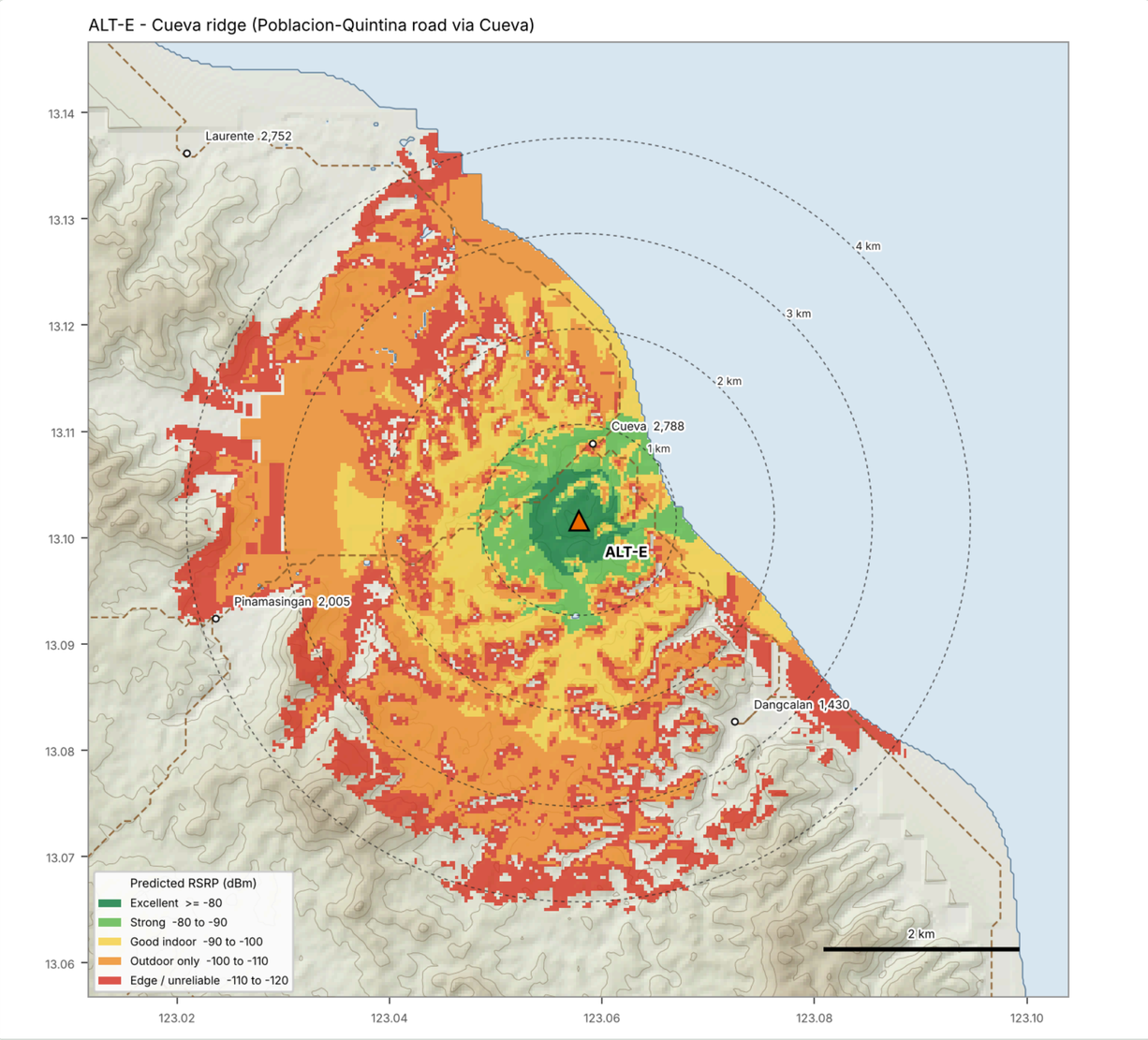


The best single site. A 153 m ridge above Malaking Ilog and Mapanique, serving the southern belt. Needs a 15 km link: specify 7/8 GHz.

LOCATION	`13.0075 N, 123.0556 E`
GROUND / ANTENNA TOP	153 m / 193 m above sea
ROAD ACCESS	985 m spur · inland Pinamasingan-San Pedro-Mapanique road
BACKHAUL	15.0 km to Bolod · 35 m masts at 15 GHz (40 m at 7 GHz)
NEW RESIDENTS	**4,314** usable · 2,323 indoor-grade
OVERLAP WITH BOLOD	0%
BARANGAYS LIFTED	Malaking Ilog 99%, Mapanique 64%, Ki-Romero 62%, San Jose 16%

BETTER COMPROMISE

ALT-E · Cueva ridge

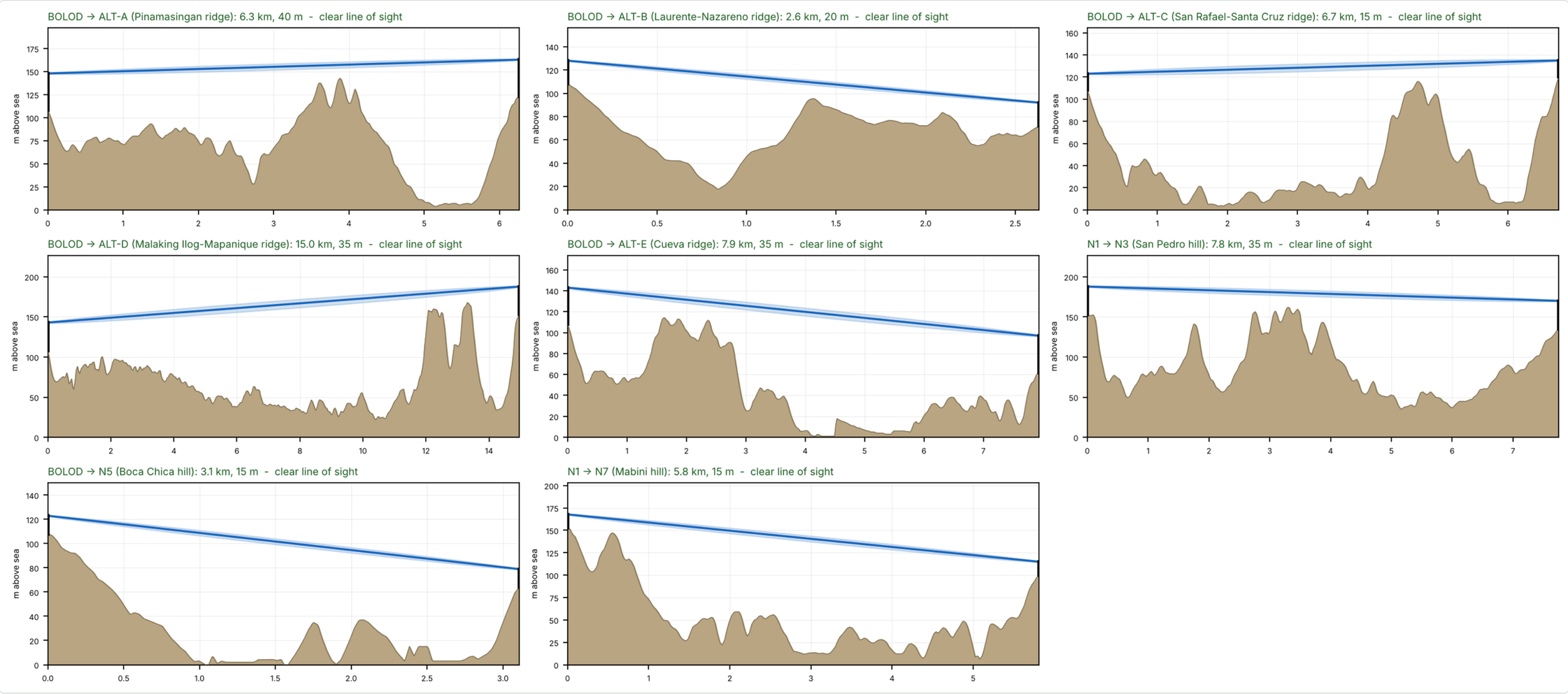


A 62 m spur on the Poblacion–Quintina road above Cueva, the third-largest unserved barangay.

LOCATION	13.1017 N, 123.0578 E
GROUND / ANTENNA TOP	62 m / 102 m above sea
ROAD ACCESS	410 m spur · Poblacion-Quintina road via Cueva
BACKHAUL	7.9 km to Bolod · 35 m masts at 15 GHz (35 m at 7 GHz)
NEW RESIDENTS	**2,813** usable · 2,069 indoor-grade
OVERLAP WITH BOLOD	0%
BARANGAYS LIFTED	Cueva 91%, Pinamasingan 12%

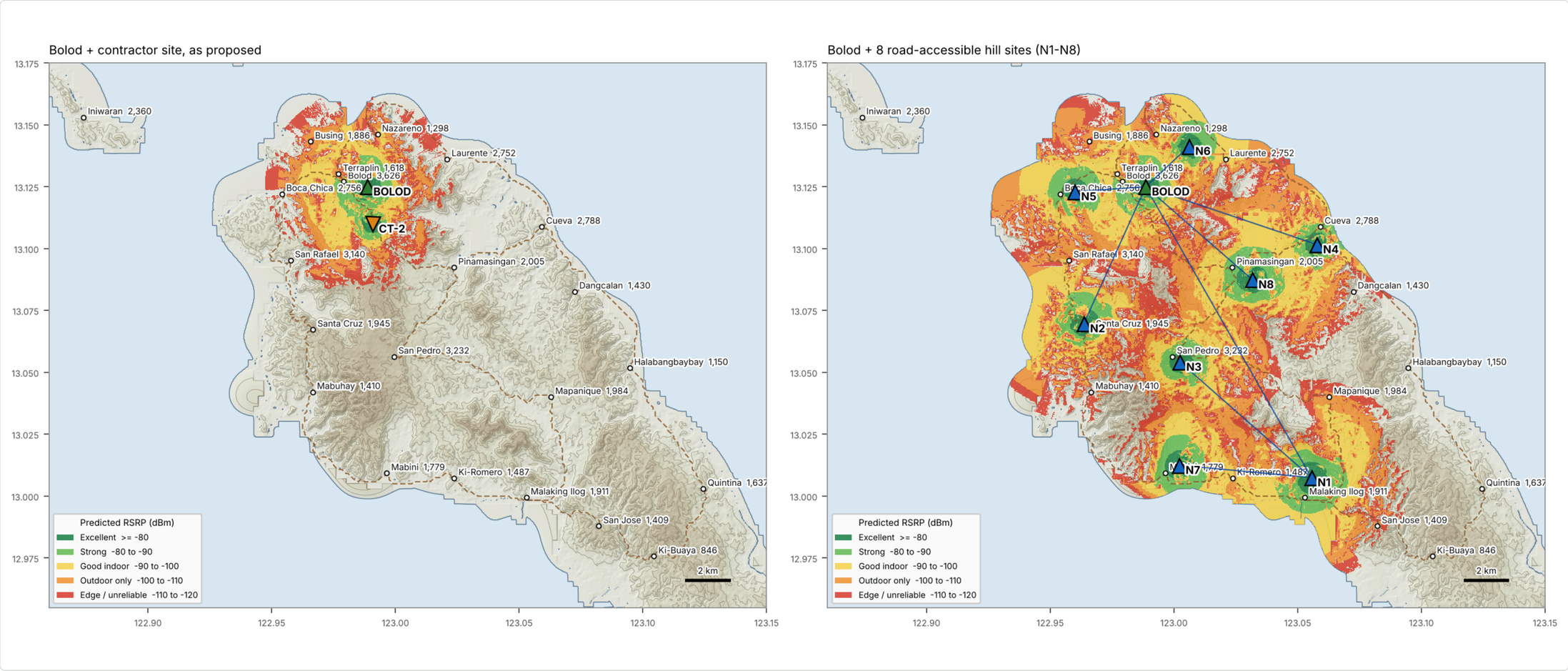
BETTER COMPROMISE

Every proposed site has a clear microwave link



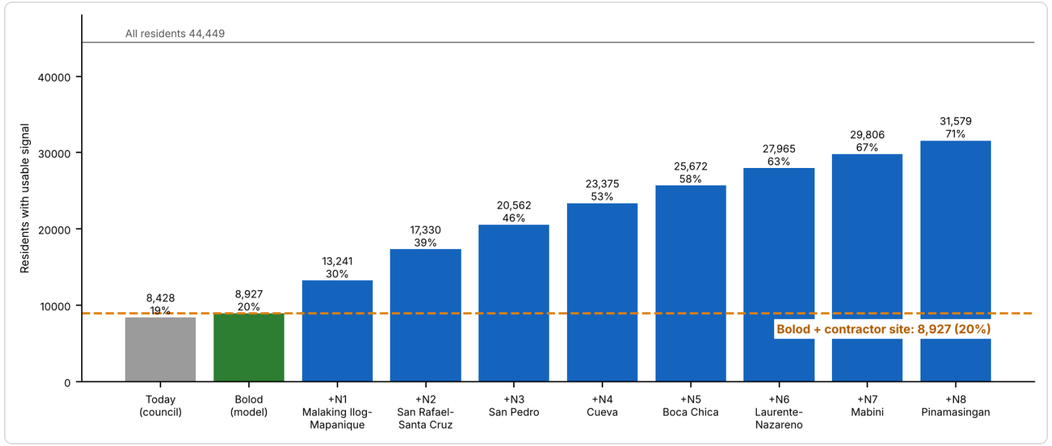
Shaded band: 60% of the first Fresnel zone at 15 GHz, the standard clearance for a licensed link. Masts are the minimum equal height at both ends. Bottom row: build-out sites that are not among ALT-A to ALT-E. N3 and N7 hop via N1 rather than directly to Bolod.

One tower can't cover a 3–4 km-reach island. A plan can.



Left: Bolod plus the contractor's site. Right: Bolod plus eight road-accessible hill sites chosen one at a time for the most new residents. Blue lines are the microwave links.

Build-out order: residents with usable signal



#	SITE	LAT, LON	GROUND	SPUR	BACKHAUL VIA	ADDS	TOTAL
N1	Malaking Ilog-Mapanique ridge (ALT-D)	13.0075, 123.0556	153 m	985 m	BOLOD · 15.0 km	+4,314	13,241 (30%)
N2	San Rafael-Santa Cruz ridge (ALT-C)	13.0697, 122.9636	120 m	43 m	BOLOD · 6.7 km	+4,089	17,330 (39%)
N3	San Pedro hill	13.0542, 123.0025	135 m	206 m	N1 · 7.8 km	+3,232	20,562 (46%)
N4	Cueva ridge (ALT-E)	13.1017, 123.0578	62 m	410 m	BOLOD · 7.9 km	+2,813	23,375 (53%)
N5	Boca Chica hill	13.1231, 122.9600	64 m	358 m	BOLOD · 3.1 km	+2,297	25,672 (58%)
N6	Laurente-Nazareno ridge (ALT-B)	13.1414, 123.0061	72 m	971 m	BOLOD · 2.6 km	+2,293	27,965 (63%)
N7	Mabini hill	13.0125, 123.0022	100 m	707 m	N1 · 5.8 km	+1,841	29,806 (67%)
N8	Pinamasingan ridge (ALT-A)	13.0875, 123.0317	123 m	723 m	BOLOD · 6.3 km	+1,773	31,579 (71%)

Phase 1 (N1 + N2) more than doubles coverage: 20% → 39%. The full eight sites reach 71%. Bolod plus the contractor's site stays at 20%.

DECISION MATRIX

Four metrics, every option

SITE	POPULATION COVERED NEW RESIDENTS, OF 36,021	BACKHAUL TO BOLOD	POWER / ROAD ACCESS	COMMERCIAL VIABILITY INDICATIVE REVENUE	VERDICT
CT Contractor, deeper Bolod	●○○○○ 0 new	●●●●● 1–2 km	●●●●● on the road, near the grid	●○○○○ no new subscribers	**Reject as a coverage project**
ALT-A Pinamasingan ridge	●●●○○ 2,615	●●●●○ 6.3 km, 40 m	●●●○○ 723 m spur	●●●○○ ₱1.6–2.4 M/yr	Minimum compromise on the contractor's road
ALT-B Laurente-Nazareno ridge	●●●○○ 2,293	●●●●● 2.6 km, 20 m	●●●○○ 971 m spur	●●●○○ ₱1.4–2.1 M/yr	Laurente fix, cheap hop
ALT-C San Rafael-Santa Cruz ridge	●●●●● 4,089	●●●●● 6.7 km, 15 m	●●●●● 43 m spur	●●●●● ₱2.5–3.7 M/yr	Phase 1
ALT-D Malaking Ilog-Mapanique ridge	●●●●● 4,314	●●●○○ 15.0 km, 35 m	●●●○○ 985 m spur	●●●●○ ₱2.6–3.9 M/yr	Phase 1, best single site
ALT-E Cueva ridge	●●●○○ 2,813	●●●●○ 7.9 km, 35 m	●●●●○ 410 m spur	●●●○○ ₱1.7–2.5 M/yr	Phase 2

Scores 1 (poor) to 5 (excellent). Revenue assumes ₱600–900 per newly covered resident per year (rural Bicol prepaid mix); replace with the operator's own figures. A 40 m greenfield with microwave typically costs ₱10–15 M, plus ₱3–6 M for a hilltop access spur and solar-hybrid power.

DECISION MATRIX

The commercial picture, read honestly

Contractor's site

No new subscribers. All revenue is traffic moved off Bolod. That justifies a capacity upgrade, which co-location on the existing Bolod towers delivers without a new structure.

Single ridge site

About 4,314 new residents at best: roughly ₱2.6–3.9 M a year against ₱13–21 M capex. Payback is about 4–7 years, slower than an urban site but positive, and it opens a captive market.

Shared network

With a 3–4 km reach, full coverage needs several sites. Multi-operator sharing, DICT common-tower and connectivity programmes, and LGU help with land and access roads are what make sites N3–N8 viable.

The council's leverage is simple: offer fast permitting, land and road help for ridge sites, and ask for nothing new for a site that adds no one.

Check the council's legal footing first

Executive Order No. 32 (s. 2023) lists Sanggunian resolutions among the requirements that are no longer prerequisites for shared telecom tower permits. **Revised JMC No. 01 s. 2021** bars LGUs from requiring documents beyond its list.

Before the hearing

- Ask the Municipal Legal Officer and DILG provincial office whether this RONO is legally required, or a courtesy endorsement the contractor is seeking.
- If a courtesy: write the conditions into a Memorandum of Agreement the contractor signs.

Levers the LGU still holds

- Municipal and barangay land for ridge sites, and right-of-way for access spurs.
- Building-permit technical review, zoning and locational clearance.
- Fast one-stop-shop processing under EO 32.
- A council resolution of findings to DICT and NTC.

HEARING STRATEGY

Five questions for the councilors to ask

Q1 **"Show us your RSRP propagation map, for your site and for the existing Bolod towers, and the difference."**

Ask for −100 and −110 dBm contours, the model and terrain data, antenna height, azimuth, tilt and power. Then: which barangays, and how many residents, gain service they do not have today?

Q2 **"Your antennas will be about 46 m lower than the ones already on Bolod hill. Name one unserved barangay you will reach that Bolod cannot."**

Ask for the terrain profile to Pinamasingan, Laurente or San Rafael. Ask whether they evaluated the Pinamasingan ridge on the same road, and for their site-search ring.

Q3 **"Which operator will be on this tower, and why not co-locate on the existing Bolod towers?"**

If the purpose is a new operator or more capacity, DICT's co-location policy points to the existing structures.

Q4 **"What is your written commitment to the 18 unserved barangays, with sites and dates?"**

Ask for a rollout schedule with coordinates and on-air dates, and a drive test shared with the council within 60 days of switching on.

Q5 **"Which frequency band will the new tower use, and will it stay up when the grid is down or a typhoon hits?"**

A sub-GHz layer (700/850 MHz) reaches further than the 3–4 km seen today. Ask for battery hours, generator or solar backup, the NSCP wind rating and typhoon restoration time.

Conditions before granting no objection

Coverage extension

- The site must give RSRP ≥ −100 dBm to at least [2,000] residents of unserved barangays not reached from Bolod. ALT-A to ALT-E pre-qualify, subject to survey.
- If they insist on deeper Bolod: capacity only, tied to a ridge site on air within [12–18] months, or the endorsement lapses.
- For a new tenant or capacity: show first why co-location on Bolod is impossible.

Evidence first

- Propagation study (Longley-Rice or ITU-R P.1812, terrain ≤ 30 m) for the site, Bolod and the difference.
- Band plan, including a sub-GHz layer.
- Microwave or fibre plan, ≥ 99.99% availability for rain zone P.
- Power plan with ≥ 8 h battery; NSCP 2015 structure; EMF compliance.

After construction

- Drive test of all 22 barangays within 60 days, with raw logs to the Sanggunian.
- Annual coverage report for three years.
- Open access to other operators on published terms.
- Restoration ≤ 72 h after typhoon damage.
- Lapses if construction has not started within 12 months.

What the LGU offers

- Priority one-stop-shop processing.
- Help finding municipal or barangay land on the chosen ridge, and spur right-of-way.
- Power-line coordination along the Pinamasingan and Poblacion–Quintina roads.
- A joint letter to DICT supporting the ridge sites.

Showing the proof in the session hall

Google Earth Pro

1. Open `results/san_pascual_review.kmz`.
2. Tools → Options → 3D View: elevation exaggeration 2.
3. Toggle "BOLOD" + "CT-2" coverage, then "BOLOD" + "ALT-A".
4. For an independent check: placemark at a site, altitude 40 m relative to ground, right-click → Show Viewshed.
5. Draw a path CT-2 → Pinamasingan → Show Elevation Profile to show the spur.

HeyWhatsThat

1. New panorama at each site, 40 m above ground, made before the hearing.
2. Show the two visibility maps in side-by-side browser tabs.
3. Path Profiler, Bolod → ALT-D, shows the microwave link clears.

CloudRF

1. Area coverage at 1800 MHz (and 758 MHz for a low-band layer), 46 dBm, 17.5 dBi, 40 m mast, 1.5 m receiver.
2. Model Longley-Rice, rural vegetated, 6 km radius.
3. Legend at −80 / −90 / −100 / −110 / −120 dBm, the classes on these slides.
4. Export KMZ into Google Earth. This is the same map the contractor should be producing.

METHOD

What this study can and cannot tell you

Solid

- The relative comparison. The same model and terrain apply to every site, so "contractor adds nothing, ridges add thousands" holds even if absolute ranges shift.
- Terrain blocking: the ridges are real and visible in Google Earth's own elevation profile.
- Calibrated to the council's field facts: four served barangays, no Laurente, 3–4 km reach.

To confirm

- The contractor's exact coordinates, and the surveyed Bolod tower positions.
- Actual road and power-line alignments; ours are inferred from terrain.
- A drive test to replace the calibration with measurements.
- A site walk at each ridge: SRTM is ±5–10 m, and land ownership matters.
- The operator's band, power and antenna plan, which sets the real reach.

Analysis code, data, KMZ and every figure: project repository, folders analysis/, data/, results/. Re-run with real coordinates by editing the constants at the top of analysis/contractor_review.py.

NEXT STEPS

Before and after the hearing

Before

1. Legal Officer opinion on the RONO's status under EO 32.
2. Request the contractor's exact coordinates and propagation map in writing.
3. Short drive test or phone survey: Busing, Nazareno, Laurente, Boca Chica, Pinamasingan.
4. Identify municipal or barangay land near ALT-C and ALT-D.

At and after

1. Present slides 2, 10, 11, 12 and 20; ask the five questions.
2. Offer the conditions and the LGU's help as a package.
3. Re-run this analysis on the contractor's own coordinates and data.
4. Track on-air dates and the drive-test report.